

CS691 CAPSTONE · FALL 2026 · DELIVERABLE 1

Wanderwise

Personalized Travel Recommender

One traveler, one city, one plan that actually works.



Team 4 - Sprint 0: Project Foundation

Sushma · Srija · Ali · Siddhu · Bala · Ram

Agenda

1

Team Roles & Responsibilities

Who owns what across the seven-person team

2

Problem Statement & Description

The gap in travel planning, and what we build

3

Personas

Who we design for, and their needs

4

Technologies & Algorithms

The stack, the ML ranker, and the optimizer

5

Project Schedule

Four sprints from foundation to MVP

6

Team Working Agreement

How we meet, collaborate, and stay accountable

7

MVP

The smallest genuinely usable product

8

Retrospective & Wiki

Sprint 0 review and project links

Team Roles & Responsibilities

Six members, clearly defined ownership.



Sushma

Data Collection & Preprocessing

Collects and cleans travel, attraction, and place data.



Srija

Scrum Master

Coordinates sprints, meetings, task tracking, and the Agile workflow.



Ali

Backend & AI

Backend services, preference extraction, and trip-state management.



Siddhu

Frontend Developer

The user interface and the interactive itinerary experience.



Bala

Integration, Testing & Evaluation

Integrates components, tests, and evaluates system performance.



Ram

Optimization Algorithm

The itinerary optimization algorithm and scheduling logic.

Finding places is not building a trip

Discovery tools help you find good places, but the traveler still has to make them fit together.

General discovery platforms

Yelp, TripAdvisor, blogs, Reddit

- Great at surfacing restaurants, attractions, and reviews
- Leave distance, opening hours, cost, and energy to you
- You still assemble the day yourself, and backtrack

Output: a list of individual places

Our personalized planner

One feasible itinerary for one traveler

- Uses interests, dates, budget, start point, walking tolerance
- Chooses a set of places that actually works together
- Respects distance, hours, cost, time, and effort

Output: a feasible plan for one person

Hypothesis: two travelers in the same city should receive visibly different itineraries, not the same “top-10” list.

Why an LLM alone isn't enough

What the language model does well

- Understands requests in plain language
- Extracts preferences into structured constraints
- Explains why a place was chosen or a trade-off made
- Interprets edits: “move the museum to Thursday”

What it cannot guarantee

- Current opening hours or temporary closures
- Live prices, transit disruptions, or availability
- That the information is factual, not hallucinated
- That the full day respects every time & cost constraint

LLM = language interface · **verified data + optimizer = source of truth for the itinerary**

What the project is

A web app that turns a destination and a person's tastes into a realistic, day-by-day itinerary the traveler can reshape in plain language. Three parts do three different jobs:

● Trained model

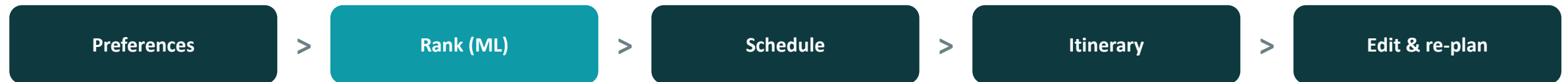
A learning-to-rank model we train scores places for this specific traveler. The heart of the project.

● Optimizer

Deterministic code packs the top picks into a schedule that respects hours, travel, budget, and pace.

● Language model

The LLM turns free text into preferences and writes the “why we picked this” explanations.



Personalization means two travelers in the same city get different plans, not the same top-10 list.

Budget-Conscious Student Traveler



Arjun Mehta

Budget-Conscious Student Traveler

Age: 24

Education: MSc Computer Science (in progress)

Hometown: Pune, India

Family: Single

Occupation: Grad student & part-time TA

"I want to explore the city without exceeding my budget or wasting time between places."

GOALS

- Stay within a fixed trip budget
- Still see the important attractions
- Cut travel time between stops

FRUSTRATIONS

- Generic plans suggest costly activities
- Recommended places are far apart
- Prices and hours are often outdated

What our app must provide: Budget-aware ranking · Verified hours & costs · Walking limit

Arjun travels between semesters on a tight budget. He researches for hours across blogs and maps but still ends up backtracking and overspending. He wants a plan that respects his budget and keeps each day efficient, so he sees more with less stress.

Parent Planning a Family Vacation



Sarah Johnson

Parent Planning a Family Vacation

Age: 38

Education: BBA, Marketing

Hometown: Chicago, USA

Family: Married, two kids (7 & 10)

Occupation: Marketing manager

"I want my family to enjoy the trip without rushing, walking too much, or missing breaks."

GOALS

- Find activities for kids and adults
- Keep each day comfortable and flexible
- Reduce travel between attractions

FRUSTRATIONS

- Generic plans pack in too many activities
- Kids get tired on long walks
- One delay derails the whole day

What our app must provide: Family-friendly filters · Meal & rest breaks · Easy re-planning

Sarah plans the family's one big trip each year, juggling two kids with different energy levels and a partner who hates rushing. She needs a realistic, flexible plan with built-in breaks that she can adjust on the fly when the day runs late.

Traveler with Limited Mobility



Michael Thompson

Traveler with Limited Mobility

Age: 63

Education: B.Ed, History

Hometown: Manchester, UK

Family: Married

Occupation: Retired secondary-school teacher

"I need to know my itinerary is physically manageable before I begin the trip."

GOALS

- Visit historical & cultural sites
- Avoid long walks and rushed days
- Get dependable accessibility details

FRUSTRATIONS

- Generic plans assume long walking ability
- Accessibility information is often missing
- Crowded schedules cause exhaustion

What our app must provide: Accessibility filters · Shorter routes & rest time · Confidence warnings

Michael loves history and museums, but a knee condition limits how far he can walk in a day. He avoids booking trips because he can't tell in advance whether a plan will be manageable. He needs walking limits respected and honest warnings when accessibility data is uncertain.

Technologies

Languages

- Python 3.11
- TypeScript

Backend & API

- FastAPI
- uvicorn
- Pydantic

Machine Learning & Data

- LightGBM (learning-to-rank)
- NumPy
- OpenStreetMap / Overpass

Frontend

- React
- Vite

LLM

- Ollama (local model)

Dev, CI & Testing

- GitHub + Actions (CI)
- uv · make
- pytest · vitest
- SQLite

Algorithms

● Learning-to-Rank

LightGBM / LambdaRank: a gradient-boosted-tree model we TRAIN to score how well each place fits a traveler.

● Ranking Evaluation

NDCG@k and Precision@k, compared against popularity, a hand-coded scorer, and an LLM-only baseline.

● Preference Extraction (NLP)

The LLM parses free text into a structured, schema-validated traveler profile.

● Itinerary Optimization

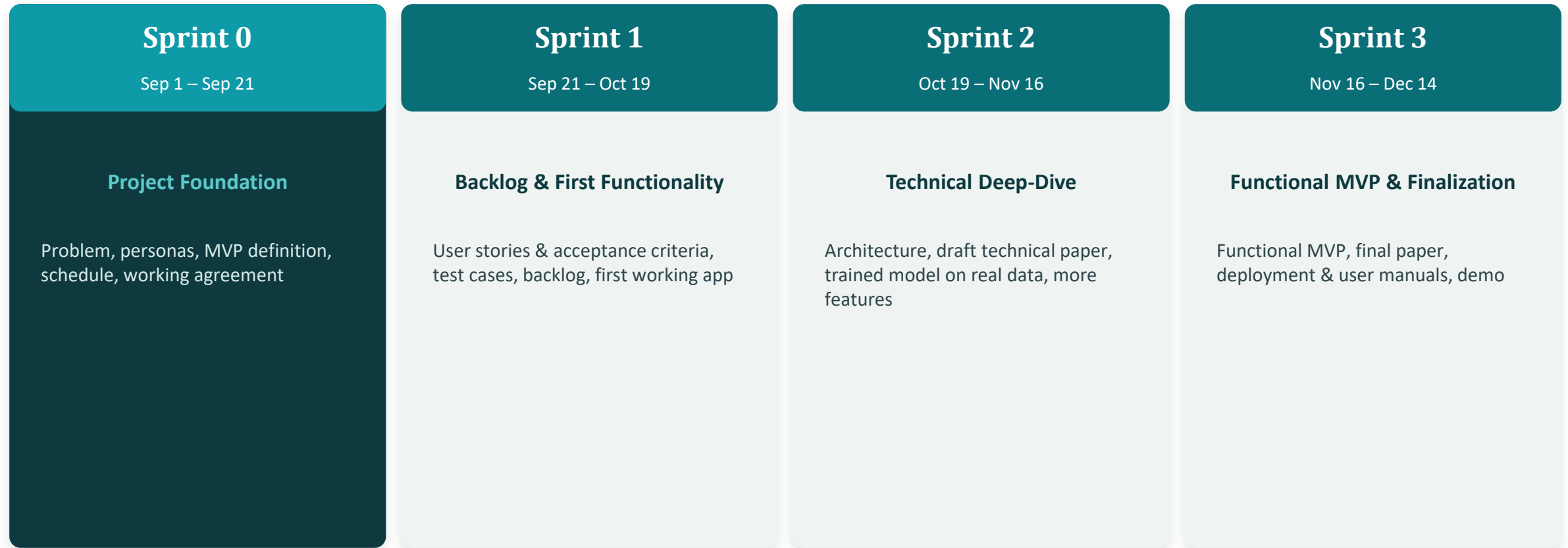
Greedy insertion evolving toward a TOPTW scheduler: packs top-ranked places under time, hours, budget & walking limits.

● Distance & Routing

Haversine estimate as a fast baseline, validated against real OSRM walking times.

Project Schedule

Four sprints from foundation to a functional MVP. Team meets weekly (in class + online), with async coordination on WhatsApp.



Each sprint ends with a working deliverable, a recorded review, a retrospective, and an updated wiki.

Minimum Viable Product

The smallest product a traveler can actually use end-to-end: One city, real data, a real plan.

In the MVP

- Enter destination, dates, interests, budget, pace
- Preferences turned into a structured traveler profile
- Candidate places retrieved for the city
- A TRAINED model ranks places for that traveler
- Optimizer builds a feasible day-by-day itinerary
- Itinerary shown with times, cost, and reasons
- Feedback adjusts the recommendations

Not in the MVP

- Login / accounts / password recovery
- Booking or payments (we recommend, not transact)
- Multi-city trips in one itinerary
- Native mobile apps

Team Working Agreement

1

Communication

WhatsApp for day-to-day updates, questions, and coordination.

2

Online Meetings

Zoom for sprint planning, progress, and problem-solving.

3

In-Person

Face-to-face when we need hands-on group work.

4

GitHub

Branch → PR → review → merge; everyone contributes.

5

Accountability

Own your tasks; commit work ≈ 48h early, not an hour before.

6

Collaboration

Support each other on integration and testing; we remind, no offense taken.

Retrospective

What went well

- Team formed and roles agreed quickly
- Repo, CI, and a working skeleton are up
- Clear, shared understanding of the design

What to improve

- Even out early workload across all six
- Lock deliverables earlier, not near deadline
- Tighten our commit / PR review rhythm

Action items for Sprint 1

- Add “definition of done” to every story
- Set a mid-sprint check-in each week

Done as a group. The two action items become tasks in the Sprint 1 backlog.

AI Disclosure

What we used it for

- Every slide with a different theme, font, layout.
- Cohesive presentation (Single color palette, font pairing, slide design).
- Persona Images

How we used it

- Formatting, design, image generation
- Reviewed and edited by each member.

Tool used: Claude and Gemini

Group Wiki Page

The wiki is the single home for every deliverable: description, team, design, personas, sprints, and artifacts.

Group Wiki

github.com/htmwf2026-team4/wiki

Code Repository

github.com/htmwf2026-team4